MILITARY SPECIFICATION

DEVICE, RESTRAINT HARNESS TAKE-UP
INERTIA-LOCKING, POWERED-RETRACTING:
GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-D-81514B(AS), dated 10 September 1969, and has been approved by the Naval Air Systems Command, Department of the Navy.

Page 3

Section 2:
Specifications:
Delete:

"MIL-S-38130	System Safety Engineering of System and Associated Subsystems and Equipments; *General Specification for*"

Standards
Insert:

"MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment; Requirements for"

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Add the following paragraph:

"3.5.2.3.1.1 Cartridge/gas generator initiation - The cartridge/gas generator initiation means shall be compatible with the system/aircraft in which it is to be used."

Paragraph 3.5.2.4: Replace first sentence with the following:

"Cable or webbing for restraint harness take-up strap/cable attachment shall conform to MIL-W-1511 or MIL-W-25361, as applicable."

FSC 1680

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Paragraph 3.5.3.1: Insert "applied through the harness take-up strap/cable" at the end of the first sentence.

Paragraph 3.5.3.2: Replace paragraph 3.5.3.2 with the following:

"3.5.3.2 Ultimate strength (static) - The PIRD, including its mounting provisions, shall withstand without structural failure a static load of 5,000 pounds applied through the harness take-up strap/cable."

Add the following paragraph:

"3.5.4 Cartridge/gas generator operation detection aids - Cartridge/gas generator operation shall produce external, long term evidence lasting and recognizable for at least thirty (30) days, that the cartridge/gas generator and, if affected, the PIRD have been operated ballistically and, therefore, are no longer suitable for use. The evidence should be tamperproof. It is desirable that the PIRD's fired condition be evident both while the PIRD is installed and also while the PIRD is not installed."

Paragraph 3.6.1.1.1:

(a) In the second sentence following "position" insert the following: "and the take-up strap/cable extended to one-half, or less, of full extension,".

(b) Following the third sentence insert the following sentence: "With the manual control handle in the "autolock" position and the take-up strap/cable extended more than one-half of full extension, the inertia-locking mechanism shall not lock for "g" loads of 1-1/2 "g's" or lower. Locking may occur for loads between 1-1/2 and 3 "g's".

(c) Add the following to the beginning of the fourth sentence (sixth sentence after incorporation of the preceding changes): "For any amount of take-up strap/cable extension from 0 through 100 percent of full strap extension,".

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Add the following paragraph:

"3.6.1.3.1 Stowage/take-up mechanism cycle life - The PIRD stowage/take-up mechanism shall have a cycle life not less than 60,000 cycles distributed as follows:

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(a) 60% 0 to 1/4 strap extension.
(b) 20% 0 to 1/2 strap extension.
(c) 10% 0 to 3/4 strap extension.
(d) 10% 0 to full strap extension."

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Paragraph 3.6.1.4.2: Replace paragraph 3.6.1.4.2 with the following:

"3.6.1.4.2 Retraction velocity -

3.6.1.4.2.1 Retraction velocity, human subjects - PIRD's within a -65°F to +75°F temperature range and conditioning herein specified shall be capable of retracting seated 3rd through 98th percentile human subjects wearing applicable configurations of flight/survival clothing/equipment without the maximum retraction velocity of the take-up strap/cable exceeding 9 ft./sec. after the first 4 inches of strap/cable retraction, for any initial extension of the personnel harness take-up strap/cable. Between +76°F and +160°F temperature range and within conditioning as herein specified, the retraction velocity shall not exceed 10 ft./sec.

3.6.1.4.2.2 Retraction velocity, human simulator - PIRD's within the temperature range and conditioning herein specified shall be capable of retracting anthropomorphic/anthropometric dummies and/or human simulators representing fully equipped 3rd through 98th percentile aircrewmen without the maximum retraction velocity of the take-up strap/cable exceeding 12 ft./sec. after the first 4 inches of strap/cable retraction, for any initial extension of the personnel harness take-up strap/cable."

Paragraph 3.6.2.1: Change "+160°F" to "+200°F."

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Paragraph 3.14: Delete "MIL-S-38130" and substitute "MIL-STD-882"

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Table I: Test No. 12 after "Crash Load" add "1/"

Table I: Add

Test Condition	Test No.	Applicable Specification	Paragraph	ABCDEFG
"Stowage/take-up mechanism cycle life	7a	81514	4.7.4.1.2	1

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Table I: Delete "Crash load" test (test 12).

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Table I: Delete tests. Nos. 22, 23 and 24 in entirety from Table. ADD:

"2/" after
High temperature
Low temperature
Normal temperature
Temperature and humidity cycling
Salt spray
Sand and dust
Vibration

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Table I: Add the following footnote:

"2/ Perform tests required under 4.7.8, 4.7.9, and 4.7.10 in accordance with requirements specified for each of the footnoted test conditions."

Paragraph 4.3.1: Delete the first sentence of the paragraph and substitute the following:

"Unless otherwise specified in the applicable detail specification, the qualification samples shall consist of:

(a) Twenty-eight PIRD's, with cartridges, of each manufacturer's part number for which qualification is desired.

(b) Sufficient quantities of cartridges, from one lot, shall be provided for obtaining Type III or Type IV, qualification, as applicable."

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Add the following paragraphs:

"4.6.2.1.1 Photographic coverage -

4.6.2.1.1.1 Motion picture coverage - High speed motion picture coverage shall be provided for all powered retraction mode tests to permit measurement of the dummy's initial and final position, velocity and acceleration during each test.

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4.6.2.1.1.1 (Continued)

At a minimum, the coverage shall include one camera positioned in front of the dummy (approximately on the extended centerline) and one to one side of the dummy (approximately 90° from the centerline). A measurement grid shall be placed to the side of the dummy opposite from, and in the field of focus of, the sideview camera. In addition, timing marks or other means shall be employed to enable data analysts to establish an accurate correlation between the photographic data and that data obtained through other means.

4.6.2.1.1.2 Still photographic coverage -

4.6.2.1.1.2.1 Pre-test photographs - Still photographs shall be taken of the installed test PIRD prior to initiation of each test herein required. Particular emphasis shall be given to photographing the installed PIRD configuration/rigging/mounting.

4.6.2.1.1.2.2 Post-test photographs - Following each test, still photographs shall be taken of any damage to the PIRD. Still photographs shall be taken throughout any post-test postmortem disassembly of the PIRD to document the condition of the parts as uncovered and as removed."

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Paragraph 4.7.3.2: In the third line delete "4,000" and substitute "5,000". In the last sentence, delete "and upon.... function normally."

Paragraph 4.7.3.3: Delete entire paragraph.

Add "4.7.4.1 Extension and retraction - "

Paragraph 4.7.4.1 renumber to 4.7.4.1.1 and change title to "Webbing extension and retraction - "

Add the following paragraph:

"4.7.4.1.2 Stowage/take-up mechanism cycle life - The PIRD stowage/take-up mechanism shall be cycled 60,000 times by pulling the take-up strap/cable to the required extension and allowing it to return to the fully retracted position. The cycle distribution shall be as follows for the full 60,000 cycles:

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(a) 2 to 1/4 strap extension.
(b) 2 to 1/2 strap extension.
(c) 2 to 1/4 strap extension.
(d) 1 to 3/4 strap extension.
(e) 2 to 1/4 strap extension.
(f) 1 to full strap extension.

This cycle test shall be completed successfully."

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Paragraph 4.7.5.1: Change "+160°F" in first sentence to "+200°F."

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Paragraph 4.7.7: Replace last two sentences of paragraph with the following:

"The device shall be tested while installed on the standard test fixture shown in Figure 2."

Add "Figure 2. Standard test fixture" as shown in this amendment.

Paragraph 4.7.7.1: Delete second sentence of paragraph.

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Paragraph 4.7.8.1: Replace paragraph 4.7.8.1 with the following:

"4.7.8.1 Methods. Means shall be provided for extending and retracting the restraint harness take-up strap/cable and obtaining an accurate calibration to the "g" load at which the device locks to permit determining whether the locking occurs within the tolerances specified in 3.6.1.1.1. This test shall be conducted for various strap/cable extensions from 0 through 100 percent of full extension. For each strap/cable extension, this test shall be accomplished by starting with the PIRD in a static condition, then increasing the "g's" from zero to the limit specified in 3.6.1.1.1, at which locking shall not occur and recording and results. The device then shall be returned to the static condition and the "g's" increased from zero to 3 "g's" and the results recorded. A description of the testing machine shall be furnished to the cognizant Government activity and that activity's approval obtained prior to conducting these tests."

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TABLE VI: Under column headed "Dummy percentile" opposite Load Condition (c),

delete " 98 " and substitute 3
3 and 98 3 and 98
3 98

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Paragraph 4.7.8.2:

(a) In the 9th line after "strap/cable" insert:
", at various extensions from 0 through 100 percent of full extension,"

(b) In the 10th line, delete: "a value of 2 "g's". The 2 "g's" " and substitute the following:
"the maximum value specified in 3.6.1.1.1 for which locking shall not occur for the strap/cable extension being tested. This "g" "

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Paragraph 4.7.10: Add the following sentence to the end of the paragraph:

"Measurements to verify that the control handle "auto-lock" and "Manual lock" position angles comply with Figure 1 shall be made during the first time the test is conducted, the last time the test is conducted and at such time as a control mechanism malfunction/failure is suspected or experienced."

Preparing activity:
Navy - AS
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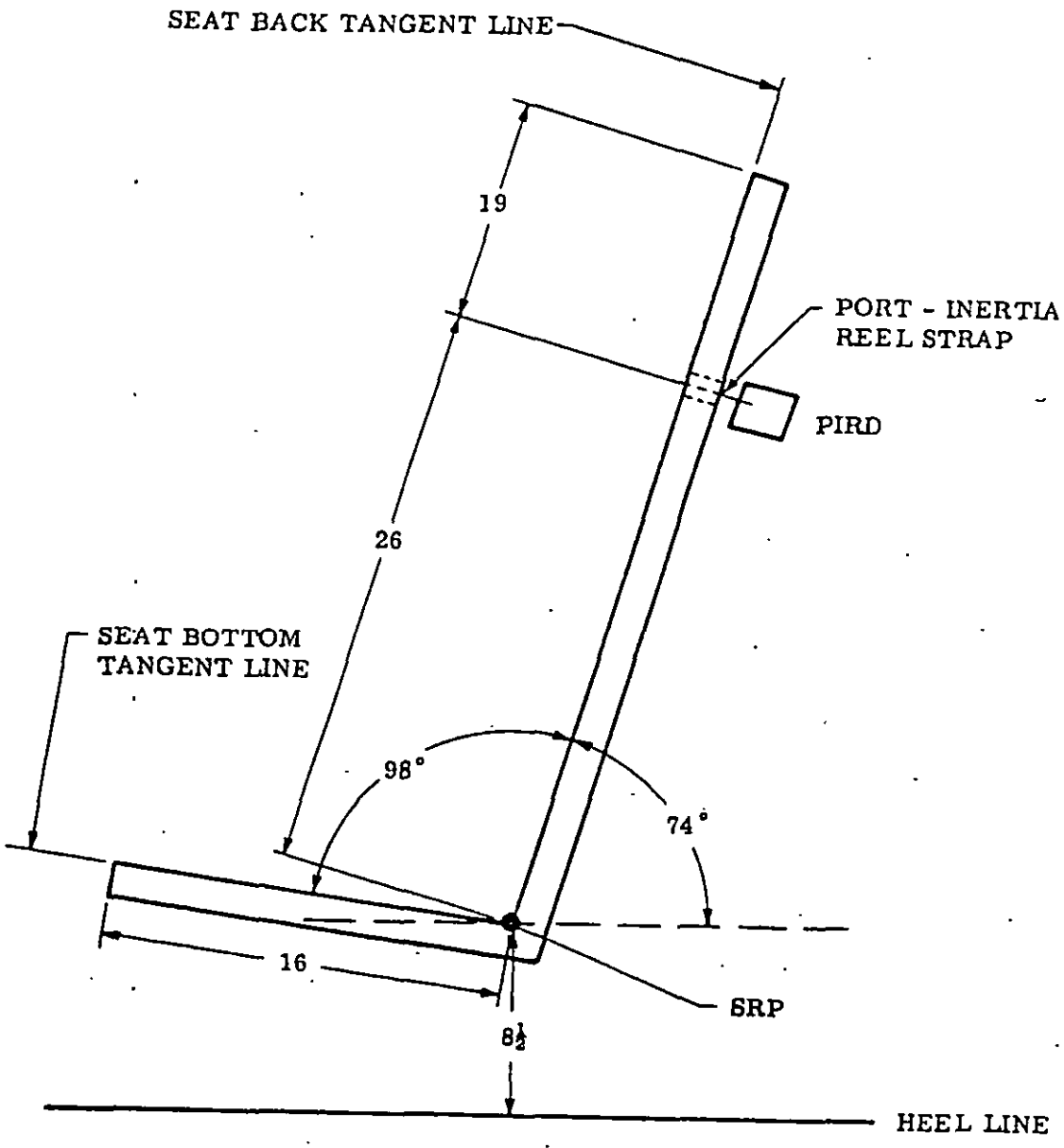


Figure 2. Standard test fixture

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